Nathan Potraz

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EXPERIENCE

Hardware Specialist & Small Business Owner
Self-Employed — June 2022 - Dec 2025, Oceanside, CA

- Engineered and fabricated over 100 custom-tuned competitive gaming controllers, earning an average 5/5 rating across 76 customer reviews.
- Developed and implemented intricate modification protocols to enable advanced features (e.g stick-profile tuning and button remapping) on custom hardware, demonstrating complex system integration.
- Optimized device hardware via precision shell modifications to achieve specific input values out of a total of 352 values - critical for high-level competitive play.

PROJECTS

Devlog
devlog.ntpotraz.dev/ • October 2025 - Present

- Architected a **high-speed Go backend** with **SQLite** and Turso for sub-1 second database performance.
- Integrated Clerk for **secure user authorization**, supporting GitHub/Google SSO and scalable login systems.
- Streamlined deployment using **Docker and GitHub Actions** for automated CI/CD to **Google Cloud Run**.
- Designed a terminal-themed React frontend with **Tailwind CSS** to enhance the developer user experience.

Japanese-English Translator
github.com/nthpotraz/cs421-translator/ • March 2024 - May 2024

- Engineered a **C++ natural language processor** using a **DFA Scanner** to tokenize complex linguistic inputs.
- Built a **Recursive Descent Parser** to enforce strict grammar rules, mirroring simulation logic structures.
- Developed a translation engine using **Hash Maps** for efficient dictionary lookup and token reordering.
- Integrated **File I/O streams** with **custom error handling** to ensure high-fidelity output and debugging.

Hack Compiler
github.com/ntpotraz/VMTranslator/ • github.com/ntpotraz/Assembler/ • March 2020 - April 2020

- Engineered a **two-pass symbolic Assembler** in **Java** to handle symbol resolution and forward referencing.
- Built a **VM Translator** converting **stack-based code into assembly** via complex register manipulation.
- Developed **branching logic** using **unique label generation** to ensure **control flow integrity** in translation.
- Designed a **Parser** to identify instruction types and a mapper to translate mnemonics into **16-bit opcodes**.

Custom DNS Resolver Simulation
github.com/Saltair1/NetworkPythonProject/ • October 2023 - November 2023

- Developed a **custom Python protocol over UDP** to simulate **Domain Name System (DNS)** infrastructure.
- Engineered **local caching logic** to validate record freshness by comparing timestamps against **TTL expiration**.
- Integrated **Pandas** to visualize resource record tables, providing **real-time monitoring** of the resolver state.
- Automated **server configuration** using JSON files to manage static records across multiple **authoritative nodes**.

EDUCATION

B.S in Computer Science
California State San Marcos • San Marcos, CA • 3.7 GPA

Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Computer Security, AI, Operating Systems, AGILE/SCRUM.

SKILLS

Languages: Golang, Python, Java, C++, TypeScript, React, SQL
Tools: Linux, Docker, CI/CD, Git, Linux, Agile, AWS, GCP